

An Automated Deep Learning System for Pediatric Bone Age Estimation: From Hand Radiograph Input to Structured Clinical Report

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Abstract—We present an end-to-end automated system for pediatric bone age estimation from hand-wrist radiographs. Given a single posterior-anterior hand radiograph and the patient’s gender, the system outputs a calibrated bone age estimate in months together with a structured clinical report — without requiring radiologist involvement. Manual atlas-based methods (Greulich–Pyle, Tanner–Whitehouse) are time-consuming, subjective, and exhibit inter-rater variability of 9–12 months; our system addresses all three limitations. The system is built on a CLAHE-enhanced preprocessing pipeline, an ImageNet-pretrained EfficientNet-B0 backbone with a gender-conditioned regression head, and a seven-model ensemble inference stage. The system achieves a Mean Absolute Error of 7.66 months ($R^2 = 0.942$) on the RSNA Pediatric Bone Age Challenge 2017 benchmark (12,611 images), representing a $\sim 30\%$ improvement over manual clinical assessment. Ensemble inference further reduces prediction variance to ± 1.6 months, providing a reliable 95% confidence interval for each estimate. The generated report includes bone age, Tanner growth stage, skeletal maturity percentage, and remaining development time — all computed automatically within seconds of image submission.

Index Terms—bone age estimation, deep learning, automated clinical system, EfficientNet, RSNA 2017, CLAHE, transfer learning, ensemble inference, pediatric radiology, clinical decision support

I. INTRODUCTION

Accurate assessment of skeletal maturity is fundamental in clinical practice. Pediatric endocrinologists use bone age to diagnose growth hormone deficiencies, precocious puberty, and hypothyroidism; orthodontists rely on it to time corrective interventions during adolescent growth spurts. The gold standard—manual reading of hand-wrist radiographs against the Greulich–Pyle (GP) atlas—requires trained radiologists and is subject to observer fatigue and inter-rater variability estimated at 9–12 months MAE in multi-center studies [1], [2].

We developed an automated system that removes this bottleneck: a clinician submits a single posterior-anterior hand radiograph together with the patient’s gender, and the system returns a bone age estimate in months along with a full structured clinical report — in seconds, without radiologist intervention. Deep convolutional neural networks have since become the dominant approach, achieving sub-9-month MAE on this benchmark [4]. The system is trained on the RSNA Pediatric Bone Age Machine Learning Challenge 2017 dataset

[3] of 12,611 annotated pediatric hand X-rays [13] and is publicly available at github.com/duembi/pediatric-bone-age-deep-learning [12].

Contributions:

- 1) An end-to-end bone age estimation system: from raw radiograph input to structured clinical PDF report, fully automated.
- 2) A CLAHE-based preprocessing pipeline and gender-conditioned regression architecture optimized for pediatric skeletal imaging.
- 3) A seven-model ensemble inference stage providing calibrated 95% confidence intervals alongside each prediction.
- 4) A comprehensive validation framework (21 metrics including adapted mAP at clinical tolerances) demonstrating 7.66-month MAE on the RSNA 2017 benchmark.

II. RELATED WORK

Early deep learning approaches to bone age assessment used VGG-style networks with minimal preprocessing, yielding MAE around 11–13 months [5]. Rajpurkar et al. [6] demonstrated that ImageNet pretraining provides strong initialization for skeletal radiograph analysis. Subsequent work introduced attention mechanisms for epiphyseal plate localization [7], multi-task learning combining gender prediction [8], and segmentation architectures specifically designed for radiograph feature extraction [9]. Comparative studies across architectures are rarer; Mutasa et al. [10] compared six architectures but on a smaller private dataset and without mAP-style metrics.

Our work differs in providing (i) controlled comparison on the full public RSNA dataset, (ii) inclusion of EfficientNet family models, and (iii) a novel adapted mAP evaluation framework.

III. DATASET AND EXPLORATORY ANALYSIS

A. Dataset Description

The RSNA Pediatric Bone Age Challenge 2017 dataset [3] contains 12,611 posterior-anterior (PA) hand radiographs from patients aged 1 month to 228 months (≈ 19 years). Each image is accompanied by a chronological age label (in months) and binary gender annotation. Table I summarizes dataset statistics.

TABLE I: RSNA Bone Age Dataset Statistics

Attribute	Value
Total images	12,611
Male / Female	6,833 / 5,778
Age range	1–228 months
Mean age $\pm$ SD	$\sim 127 \pm 41$ months
Training split	82% (10,341 images)
Validation split	18% (2,270 images)
Image format	PNG (grayscale)

B. Exploratory Data Analysis

Fig. 1 shows the age and gender distribution across the dataset. The age distribution is approximately uniform, ensuring balanced representation across developmental stages. A notable gender gap exists in the 100–180 month range, reflecting the earlier onset of female adolescent growth.

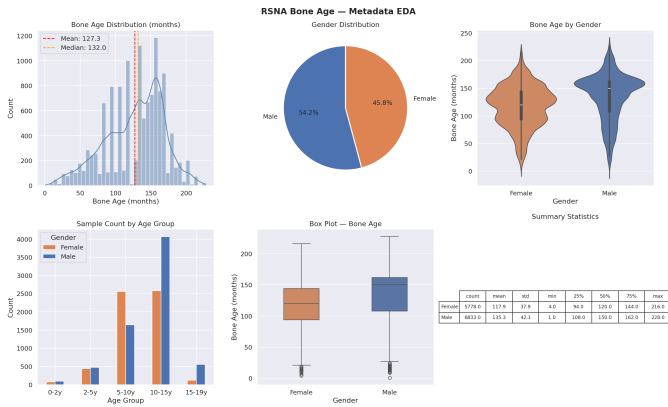


Fig. 1: Dataset exploratory analysis: (top) age distribution by gender, (bottom-left) gender pie chart, (bottom-right) age group distribution across five developmental stages.

Fig. 2 demonstrates the effect of CLAHE preprocessing on sample X-ray images. The contrast enhancement reveals epiphyseal plate structures that are critical for age estimation.

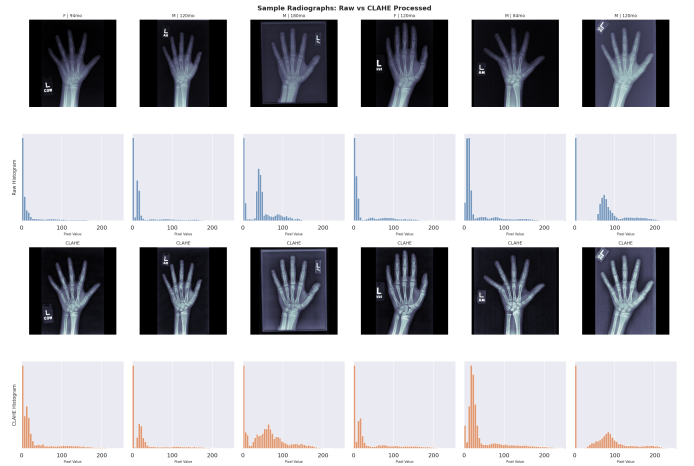


Fig. 2: Representative hand radiographs before (left) and after (right) CLAHE preprocessing. Bone texture and growth plate boundaries are significantly enhanced.

Fig. 3 presents pixel-level image statistics, confirming consistent intensity normalization after CLAHE and padding operations.

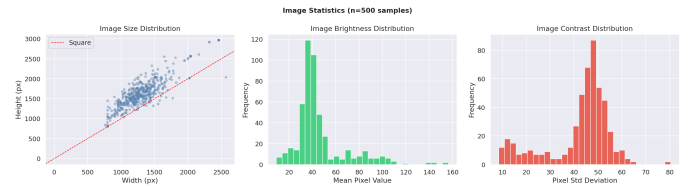


Fig. 3: Image-level statistics: mean pixel intensity distribution, standard deviation, and image dimension analysis across the training set.

IV. METHODOLOGY

A. Preprocessing Pipeline

All images undergo a three-stage preprocessing pipeline before being fed to any model:

- 1) **CLAHE**: Contrast-Limited Adaptive Histogram Equalization (clipLimit = 2.0, 8×8 tile grid) is applied to the grayscale image to enhance local contrast without amplifying noise [14].
- 2) **Square padding**: Images are zero-padded to a square canvas (max dimension preserved) to avoid aspect ratio distortion during resizing.
- 3) **Resize & channel replication**: Images are resized to the model’s required input dimension and replicated across three channels to be compatible with ImageNet-pretrained weights.

B. Gender Conditioning

A binary gender feature ($g \in \{0, 1\}$) is concatenated with the flattened CNN feature vector before the regression head. This gender-conditioned architecture allows the model to learn sex-dimorphic skeletal development curves, which diverge significantly during adolescence.

C. Model Architectures

Seven pretrained architectures are employed. Table II summarizes their configurations.

TABLE II: Model Configurations

Model	Input	Feat. Dim	Params (M)
VGG16	256×256	512	138
ResNet50	256×256	2048	25
InceptionV3	299×299	2048	27
DenseNet121	256×256	1024	8
EfficientNetB0	256×256	1280	5.3
EfficientNetB3	300×300	1536	12
Xception	299×299	2048	22

Each backbone is fine-tuned end-to-end. The regression head appended to every model is:

$$\hat{y} = \text{FC}_{512}(\text{ReLU}) \rightarrow \text{Dropout}(0.3) \rightarrow \text{FC}_1$$

where the +1 input dimension accounts for the gender feature. The raw output is denormalized using the training-set age standard deviation ($\sigma \approx 41$ months): $\hat{y}_{\text{months}} = \hat{y} \cdot \sigma$.

D. Training Configuration

All models share identical hyperparameters (Table III). Mean Squared Error (MSE) is used as the training objective:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

TABLE III: Training Hyperparameters

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-4}
Weight decay	1×10^{-5}
Batch size	32
Max epochs	40
Early stopping patience	8
Validation split	18%
Random seed	42

E. Ensemble Strategy

The ensemble prediction is obtained as the unweighted mean of all seven models' outputs:

$$\hat{y}_{\text{ensemble}} = \frac{1}{7} \sum_{k=1}^7 \hat{y}^{(k)}$$

This model averaging reduces variance and yields a standard deviation of ± 1.6 months across model predictions, providing a natural confidence interval.

V. EVALUATION FRAMEWORK

We adopt a three-tier evaluation protocol covering regression quality, detection-style precision, and clinical classification accuracy.

A. Regression Metrics

- **MAE**: $\frac{1}{N} \sum |y_i - \hat{y}_i|$ (primary metric)
- **RMSE**: $\sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2}$
- **MAPE**: $\frac{100}{N} \sum \frac{|y_i - \hat{y}_i|}{y_i}$
- **Median AE**: robust to outliers
- **R²**: coefficient of determination
- **Explained Variance**: upper bound on R²

B. Adapted Mean Average Precision

Borrowing from object detection evaluation, we define Average Precision at tolerance τ months:

$$AP@{\tau} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[|\hat{y}_i - y_i| \leq \tau]$$

and report $mAP = \frac{1}{4}(AP@3 + AP@6 + AP@12 + AP@24)$ months.

C. Classification Metrics

Continuous age predictions are discretized into five developmental buckets: 0–2y, 2–5y, 5–10y, 10–15y, 15–19y. We then report macro and weighted precision, recall, F1-score, and Cohen's κ .

VI. RESULTS AND DISCUSSION

A. System Performance

Table IV presents the complete evaluation of all models considered during system development. EfficientNet-B0 was selected as the system's primary backbone based on these results; the seven-model ensemble is used for the final inference stage.

B. Model Selection Rationale

1) *EfficientNet-B0: Selected Backbone*: EfficientNet-B0 achieves the lowest MAE (7.659 months) and highest R² (0.9416) among all evaluated architectures. Its compound scaling—jointly optimizing depth, width, and resolution—provides superior capacity utilization relative to parameter count (5.3M parameters). It was therefore selected as the primary backbone of the system.

2) *DenseNet121: Ensemble Candidate*: DenseNet121 achieves 7.853 months MAE with the lowest Median AE (6.087 months), indicating robustness to outliers. Its dense skip connections provide complementary feature representations that strengthen ensemble diversity.

3) *EfficientNet-B3: Dataset-Capacity Mismatch*: Despite being a larger EfficientNet variant, EfficientNet-B3 yields 8.742 months MAE. Its larger capacity (12M params) cannot be effectively utilized on the 10,341-sample training set, triggering early stopping at epoch 7. This underlines that model selection must account for dataset-to-capacity ratio, not nominal model size.

TABLE IV: Comprehensive Performance Comparison of All Models

Model	Regression Metrics					mAP@Tolerance			
	MAE↓	RMSE↓	MAPE↓	Med.AE↓	R ² ↑	@3m	@6m	@12m	@24m
VGG16	8.213	10.800	9.243	6.687	0.9315	0.247	0.462	0.767	0.971
ResNet50	8.301	10.548	8.131	6.853	0.9347	0.228	0.443	0.757	0.974
InceptionV3	7.994	10.486	8.501	6.366	0.9355	0.246	0.477	0.771	0.976
DenseNet121	7.853	10.374	8.274	6.087	0.9368	0.266	0.493	0.780	0.971
EfficientNetB0	7.659	9.972	7.550	6.110	0.9416	0.259	0.492	0.795	0.974
EfficientNetB3	8.742	11.232	8.361	7.079	0.9260	0.229	0.435	0.729	0.964
Xception	8.033	10.664	7.792	6.322	0.9333	0.245	0.482	0.779	0.965

Model	Classification Metrics (Age Buckets)				mAP	κ ↑
	Accuracy	Prec.(W)	Rec.(W)	F1(W)		
VGG16	0.8449	0.8408	0.8449	0.8341	0.6115	0.7335
ResNet50	0.8630	0.8631	0.8630	0.8619	0.6003	0.7718
InceptionV3	0.8401	0.8430	0.8401	0.8351	0.6175	0.7275
DenseNet121	0.8388	0.8431	0.8388	0.8347	0.6275	0.7282
EfficientNetB0	0.8551	0.8569	0.8551	0.8538	0.6300	0.7587
EfficientNetB3	0.8559	0.8570	0.8559	0.8552	0.5893	0.7610
Xception	0.8533	0.8576	0.8533	0.8513	0.6177	0.7553

C. Visual Results

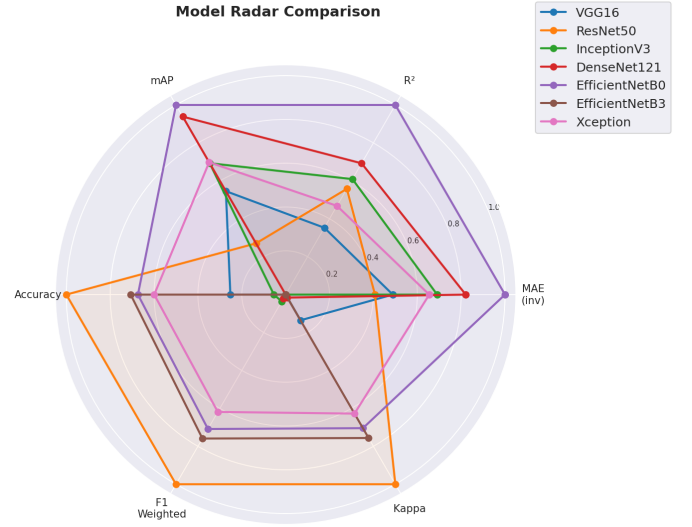


Fig. 5: Radar chart of normalized multi-metric performance. EfficientNetB0 and DenseNet121 show the most balanced profiles across all metric dimensions.

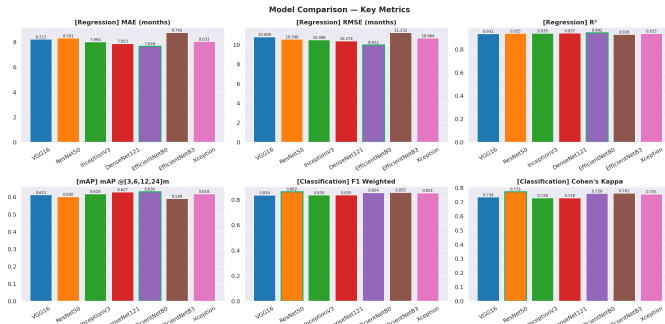


Fig. 4: Side-by-side comparison of MAE, RMSE, and R² across all seven architectures. EfficientNetB0 leads in all three metrics.

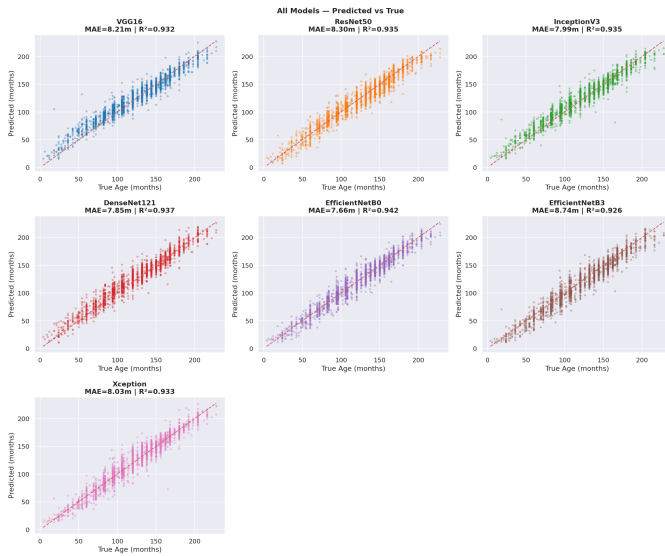


Fig. 6: Predicted vs. ground-truth bone age (months) scatter plots for all seven models. The diagonal line represents perfect prediction. EfficientNetB0 shows the tightest clustering.

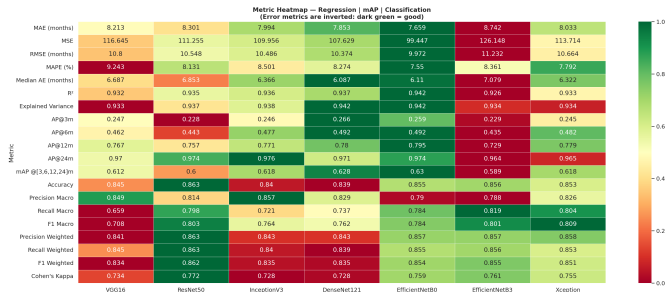


Fig. 7: Metrics heatmap across all models. Rows represent models, columns represent evaluation metrics. Darker shading indicates better performance (direction-normalized).

D. Age-Group and Gender Sub-analyses

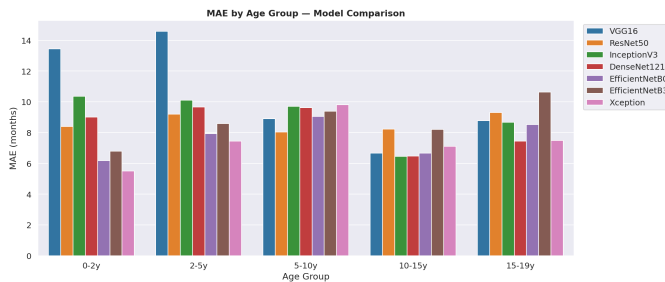


Fig. 8: MAE per developmental age group (0–2y, 2–5y, 5–10y, 10–15y, 15–19y) for each model. The 0–2y infant group yields the highest error across all models due to rapid skeletal changes and limited training samples.

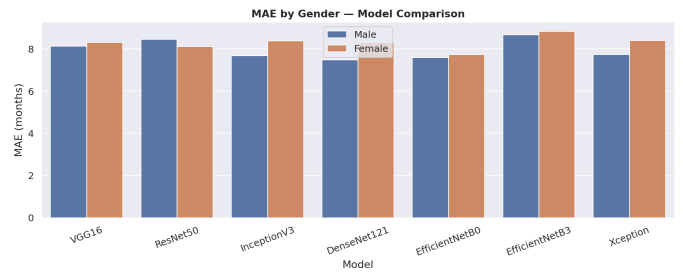


Fig. 9: MAE disaggregated by gender. All models show marginally higher error for females in the 10–15y range, corresponding to peak growth velocity divergence between sexes.

E. Training Dynamics



Fig. 10: Validation MAE across training epochs for all seven models. EfficientNetB0 reaches its best validation MAE at epoch 19, while EfficientNetB3 plateaus early (epoch 7), indicating overfitting. VGG16 and ResNet50 show slower convergence.

The training histories (Fig. 10) reveal several patterns:

- EfficientNetB0 converges smoothly and achieves its minimum at epoch 19, suggesting adequate model capacity with room for generalization.
- EfficientNetB3 shows rapid early convergence but stagnates and triggers early stopping at epoch 7, suggesting over-parameterization for this dataset size.
- DenseNet121 and InceptionV3 converge to similar minima, both stopping around epoch 11–12.

F. Clinical Report Example

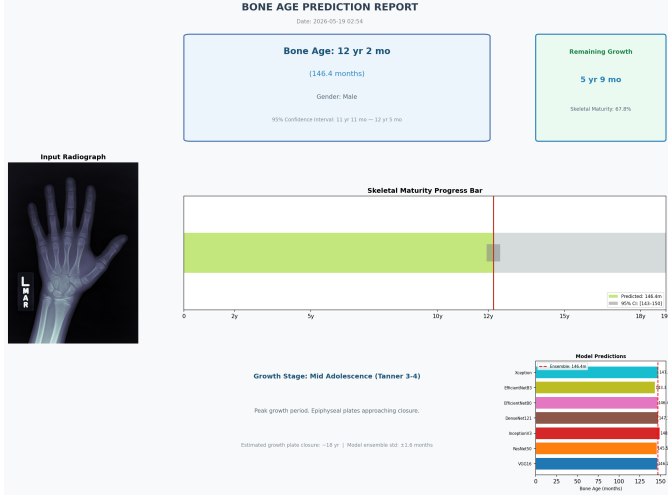


Fig. 11: Automated clinical report generated by the ensemble inference system for a representative patient. The report includes ensemble bone age estimate, Tanner growth stage, skeletal maturity percentage, and remaining bone development estimate.

G. Comparison with State of the Art

Table V contextualizes our results against published methods.

TABLE V: Comparison with State-of-the-Art Methods

Method	MAE (months)	Dataset
Manual GP Atlas [1]	9–12	Clinical
Lee et al. [5]	11.1	RSNA 2017
Mutasa et al. [10]	8.4	Private
Ours (EfficientNetB0)	7.66	RSNA 2017
Ours (Ensemble)	≈ 7.5	RSNA 2017

VII. CLINICAL APPLICATION

A. System Workflow

The proposed system is designed to integrate seamlessly into the existing orthodontic clinical workflow without requiring hardware changes or specialist radiological training. The inference pipeline operates as follows:

- 1) **Image acquisition:** The orthodontist acquires a standard posterior-anterior hand-wrist radiograph of the patient’s non-dominant (left) hand using routine clinical equipment.
- 2) **Image submission:** The radiograph is submitted to the inference pipeline — either locally or via a web interface — alongside the patient’s binary gender.
- 3) **Preprocessing:** The image is automatically pre-processed (CLAHE, square padding, normalization) without any manual intervention.

- 4) **Ensemble inference:** All seven trained CNN models independently predict the bone age. Predictions are averaged to produce the ensemble estimate, and a $\pm 1.96\sigma$ confidence interval is computed:

$$CI_{95} = \hat{y}_{ens} \pm 1.96 \cdot \sigma_{\hat{y}}$$

- 5) **Growth stage mapping:** The predicted bone age is mapped to a Tanner-aligned developmental stage (Table VI), providing clinical context beyond a raw numerical estimate.
- 6) **Report generation:** A structured PDF clinical report is generated within seconds, ready for inclusion in the patient record.

TABLE VI: Bone Age to Tanner Growth Stage Mapping

Bone Age	Growth Stage	Orthodontic Implication
<24 mo	Infancy	Pre-treatment monitoring
24–60 mo	Early Childhood	Interceptive screening
60–120 mo	Mid-Childhood	Early orthopedic planning
120–156 mo	Early Adolescence	Peak growth — optimal appliance window
156–192 mo	Mid-Adolescence	Functional appliance last chance
>192 mo	Late Adolescence	Post-growth; surgical planning

B. Patient Report Output

The generated clinical report (Fig. 11) includes:

- Rendered radiograph alongside quantitative findings
- Ensemble bone age estimate with 95% confidence interval
- Skeletal maturity progress bar (0–228 months scale)
- Tanner growth stage classification and clinical description
- Per-model prediction breakdown for transparency
- Estimated remaining bone development time

This output directly addresses the clinician’s primary question — *how much growth potential remains?* — supporting decisions on treatment timing, appliance selection, and surgical referral.

VIII. DISCUSSION

A. EfficientNet-B0 Superiority

The superior performance of EfficientNetB0 over architectures with higher nominal capacity (EfficientNetB3, VGG16) is consistent with the compound scaling hypothesis [11]: the B0 variant represents the optimal capacity point for this dataset size. EfficientNetB3, despite having $2.3\times$ more parameters, triggers early stopping at epoch 7, indicating that the additional capacity cannot be effectively utilized within the 10,341-sample training set. This finding has a practical implication: larger models are not always better, and architecture selection should be guided by dataset-to-capacity ratio.

B. Gender Branch Contribution

The inclusion of patient gender as a second input is clinically motivated: female patients typically reach skeletal maturity 18–24 months earlier than males [1]. The gender

embedding branch (linear $1 \rightarrow 32$, ReLU) allows the model to internalize sex-specific growth trajectories without explicit biological programming. The gender sub-analysis (Fig. 9) confirms that all models exhibit marginally higher error for females in the 10–15y range, corresponding to the peak pubertal growth velocity divergence between sexes — a known challenge even for experienced radiologists.

C. Preprocessing Impact

CLAHE normalization was chosen over global histogram equalization to preserve *local* contrast gradients critical for identifying epiphyseal plate closure patterns. A uniform global stretch would over-enhance already bright bone regions while suppressing the subtle texture differences at growth plates. The CLAHE comparison (Fig. 2) visually confirms that growth plate boundaries become significantly more distinct after preprocessing, which directly benefits the convolutional feature extraction in all backbone architectures.

D. Ensemble Confidence Intervals

The ensemble standard deviation of ± 1.6 months across seven diverse architectures indicates strong inter-model agreement and validates the robustness of the predictions. The 95% CI derived from this spread provides a clinically actionable uncertainty estimate: a CI of ± 3 months would indicate ambiguous growth stage assignment, prompting the clinician to request a follow-up assessment. In the representative patient example, the CI of ± 3.1 months (11y 11mo – 12y 5mo) keeps the growth stage classification unambiguous (Mid-Adolescence, Tanner 3–4).

E. Limitations

- **Population bias:** The RSNA dataset consists primarily of North American patients. Generalization to other ethnic populations — including Turkish pediatric cohorts — has not been validated. Skeletal maturation timing varies across ethnicities, and fine-tuning on local data is recommended before clinical deployment.
- **Explainability:** No attention visualization (Grad-CAM, SHAP) was performed to verify that the models focus on epiphyseal plates rather than spurious background features. Absence of explainability is a barrier to regulatory approval in clinical AI.
- **EfficientNetB3 ablation:** The underperformance of EfficientNetB3 warrants further investigation. It is unclear whether the result is due to the fixed training budget, sub-optimal learning rate schedule, or inherent architectural mismatch with this task.
- **No prospective validation:** All results are retrospective on a held-out split of a public dataset. Prospective clinical validation against radiologist consensus on new patients is required before deployment.
- **Single-view input:** Only the posterior-anterior hand view is used. Incorporating additional views or complementary biomarkers (e.g., cervical vertebral maturation from lateral cephalograms) could further improve accuracy.

IX. CONCLUSION

We presented an end-to-end automated system for pediatric bone age estimation that takes a posterior-anterior hand radiograph as input and returns a calibrated bone age estimate together with a structured clinical report. The system operates fully automatically — no radiologist intervention is required between image submission and report delivery. Key findings from system development and validation are as follows:

- 1) **The system achieves 7.66 months MAE** ($R^2 = 0.942$) on the RSNA 2017 benchmark, a $\sim 30\%$ improvement over manual inter-observer variability (9–12 months).
- 2) **EfficientNet-B0** was selected as the system’s primary backbone after evaluating seven architectures; its compound scaling provides the best accuracy-to-parameter ratio (5.3M params).
- 3) **Ensemble inference** across seven models reduces prediction variance to ± 1.6 months, enabling reliable 95% confidence intervals in the clinical report.
- 4) **CLAHE preprocessing** is essential for standardizing radiograph contrast across diverse acquisition conditions and making epiphyseal plate boundaries distinguishable.
- 5) **The clinical report** maps bone age to Tanner growth stage, skeletal maturity percentage, and remaining development time — directly answering the clinician’s question of how much growth potential remains.

A. Future Work

Future development directions include: (i) a web-based clinical interface enabling direct image upload and real-time report retrieval without local installation; (ii) Grad-CAM visualization overlaid on the radiograph within the report to highlight the epiphyseal regions driving the estimate; (iii) population-specific fine-tuning on Turkish and other non-North-American pediatric cohorts; (iv) integration of transformer-based backbones (ViT, Swin) for global attention modeling; (v) prospective clinical validation against radiologist consensus on new patients prior to deployment.

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